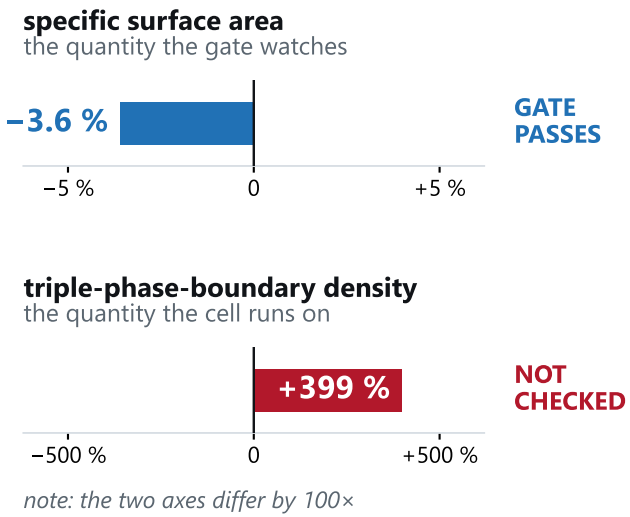
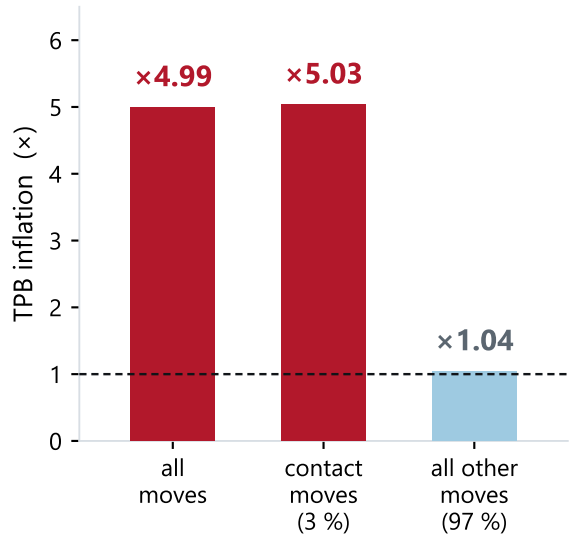
A validity check that destroys what it certifies

Voxel coarsening operators on public FIB-SEM tomograms of Ni–YSZ fuel-cell electrodes.

(a) One run, judged two ways



(b) 3 % of the moves cause all of it



WHY THE GATE DOES NOT OBJECT

80 %

of accepted moves change the area by exactly zero, so the criterion never prices them at all.

TIGHTENING IT IS NOT ENOUGH

×4.99 → ×1.53

demanding a strict area decrease shrinks the artifact. It does not remove it.

Surface area is the wrong invariant to validate a coarsening rule against.

One check costs a line and disqualified three operators here: an operator whose TPB retention exceeds 1 is roughening, not coarsening.

Pore-network analysis · Percolation · Pre-registered operators · Reproducible build

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